

Little Red Dots: Stellar Eruptions in Disguise?

Type IIIn supernovae and eta Carinae's Great Eruption offer a new model for these enigmatic objects.

arXiv:2606.30711, based on JWST observations

First author / year / source

arXiv:2606.30711 doi:10.48550/arXiv.2606.30711



Background

Little Red Dots are compact, red sources seen by JWST, often hosting active galactic nuclei. Their spectra show broad Balmer lines and a lack of variability, puzzling astronomers.

Their overmassive black hole estimates may be wrong; the true nature of their engines is debated.

What this paper is selling

This paper draws parallels with stellar eruptions, proposing that LRDs are super-Eddington, intermediate-mass engines enshrouded in dense winds. This explains their spectra and suggests lower black hole masses.

Key results to remember

- LRDs cluster at ~5000 K, like Type IIIn SNe and eta Carinae.
- Wind-shock model reproduces line profiles and Balmer breaks.
- Engine masses are likely 10^3 - 10^6 solar masses, not overmassive.

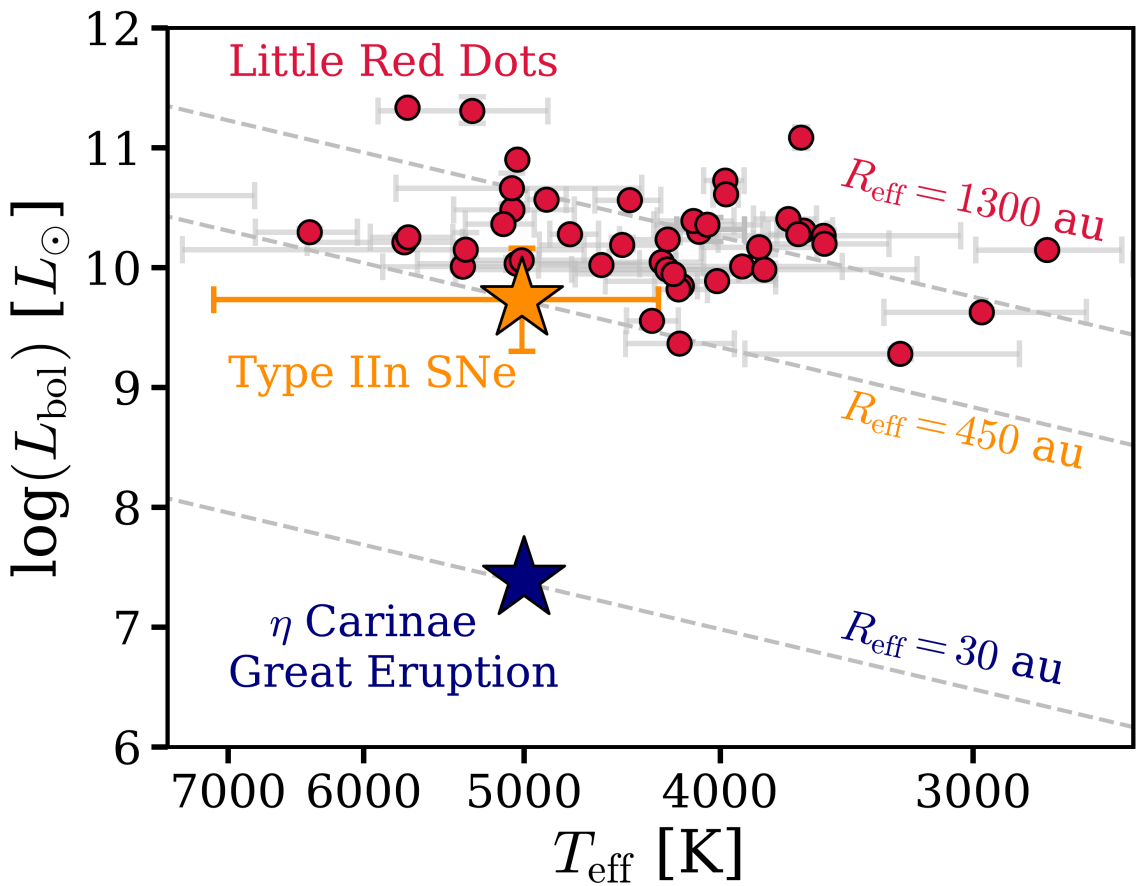


Fig. 4. LRDs, IIIn SNe, and eta Carinae share similar temperatures despite huge luminosity range.

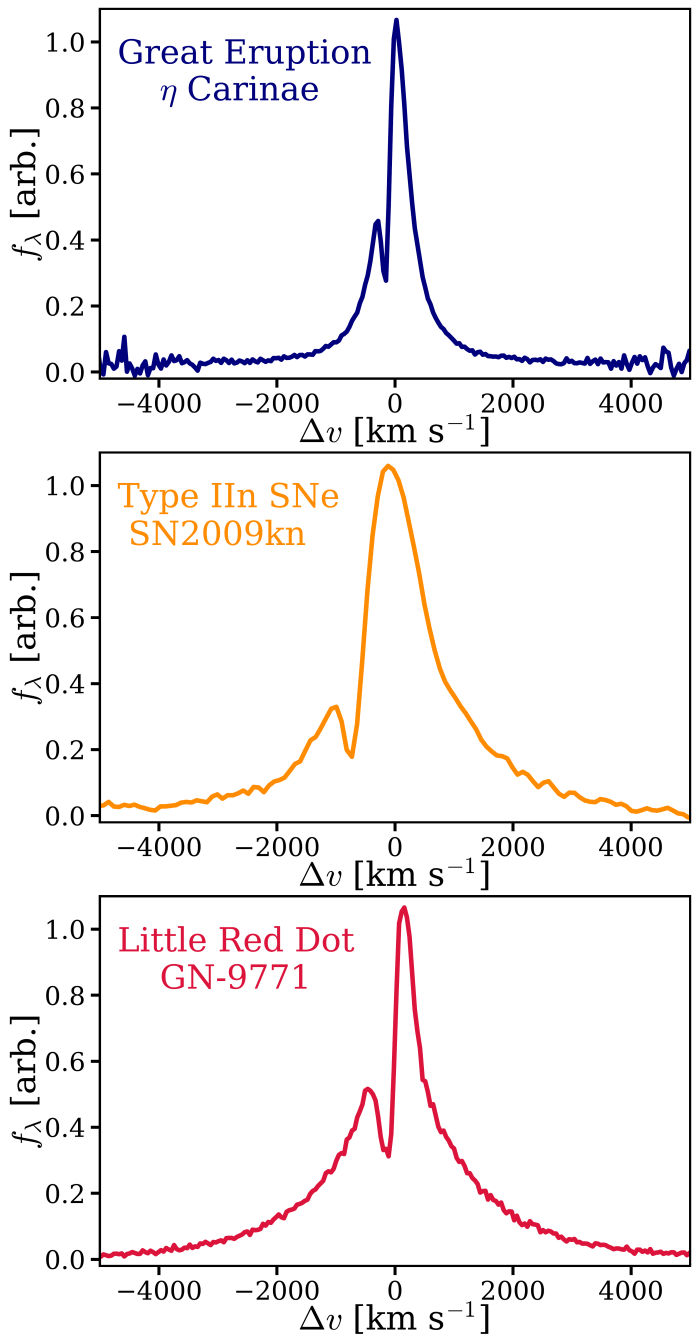


Fig. 1. Halpha profiles from eta Carinae, IIIn SNe, and LRDs show striking similarities.